BRUXELLES MOBILITÉ
BRUSSEL MOBILITEIT

SERVICE PUBLIC RÉGIONAL DE BRUXELLES
GEWESTELIJKE OVERHEIDSDIENST BRUSSEL

FOSS4G BELGIUM 2026

ceci n'est pas une conférence

… from Groene Hond to Palais Stoclet, a Geo-MCP for Brussels

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Context

- Database questions had to pass by a database specialist before
- AI is making databases accessible for more people
- To prompt a geospatial database, the spatial context of a prompt should be retrieved.

Context

Palais Stoclet



Groene Hond



DATABASE QUERY

Give me all the drains

from **Groene Hond**
to **Palais Stoclet**

SPATIAL DEFINITION



Problem

A problem arrives as words — we need to turn it into a place

1 · A PROBLEM COMES IN

- Potholes near Highroad
- Drains from Groene Hond to Palais Stoclet
- Traffic lights on Ninoofsesteenweg

→

2 · LOCATE IT

Where exactly?

Which road, address or zone? A name has to become a geometry.

→

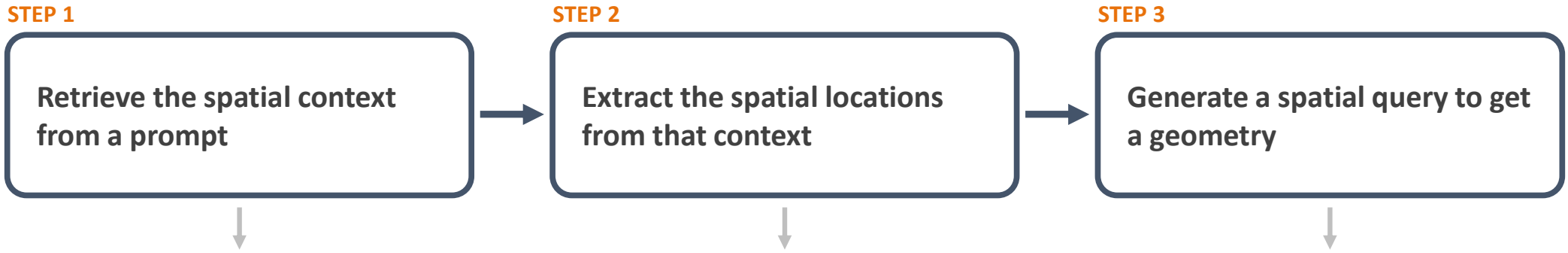
3 · ACT ON IT

Select and assign

Pick the assets (drains, potholes, lights…) on that geometry and hand them to the right team.

Step 2 is what the Geo-MCP automates.

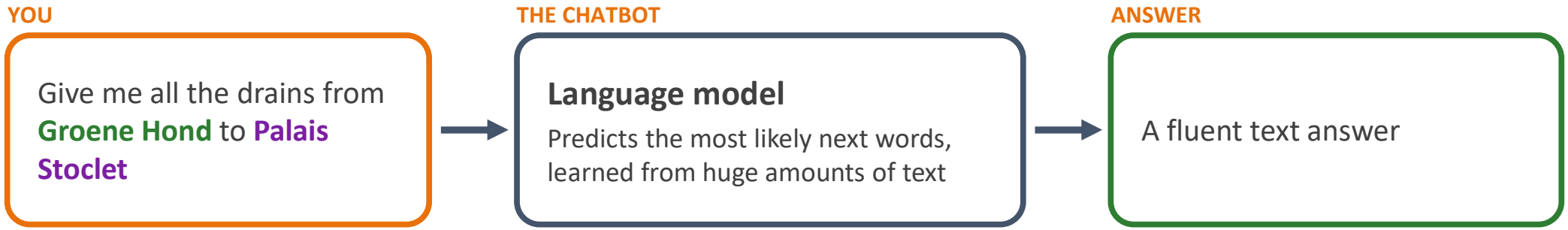
Problem



=> The way we resolve problems at Brussels Mobility

Chatbot

A chatbot is a language model with a chat window



THE LIMIT

It only knows the text it was trained on — not the streets of Brussels, and not our databases.

Solution

Ollama runs open language models on our own server

Ollama

Runs open models locally

One simple HTTP API

Our data stays in-house

Qwen, Llama, Mistral, Gemma…

Qwen2.5 · 7B

Parameters
7.6 billion

GPU memory
≈ 5 GB

Speed
~56 tok/s

Level (mid-2024)
≈ GPT-3.5

Qwen2.5 · 14B

Parameters
14.7 billion

GPU memory
≈ 10 GB

Speed
~28 tok/s

Level (mid-2024)
in between

Qwen2.5 · 32B

Parameters
32.5 billion

GPU memory
≈ 20 GB

Speed
~8 tok/s

Level (mid-2024)
≈ GPT-4o mini

CLOUD · COMPARISON

Claude Sonnet 5.5 · medium

Parameters
not disclosed
(> 1000 billion)

GPU memory
none (cloud)

Speed
not tested

Level (2026)
frontier class

Solution

A chatbot only produces text — to find a geometry it needs three things

1. **Understand the sentence**
   Which places are mentioned, and what is the spatial relation: near, on, between, from … to?

2. **Know the places**
   A catalog of Brussels streets, addresses, stops and districts — each with its geometry.

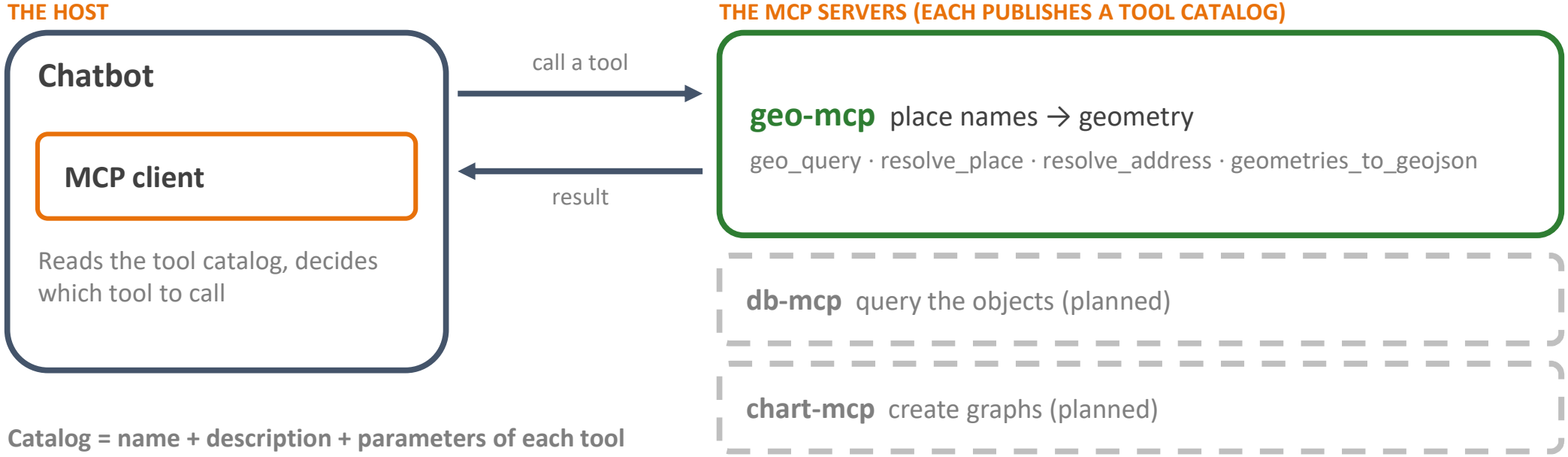
3. **Build the spatial query**
   Turn places + relation into a real SQL filter, instead of letting the model guess.

=> The chatbot needs tools it can call

Tools make a chatbot smart: a top model with bad tools is just a fluent text generator.

Solution

MCP: an open standard to plug tools into a chatbot



Catalog = name + description + parameters of each tool

Solution

Built with eval-driven development: a prompt-validation loop

1 · Prompts

> 25 test prompts in EN / FR / NL: typos, short names, tricky cases

2 · Run

Batch run through the real pipeline, every step logged

3 · Inspect

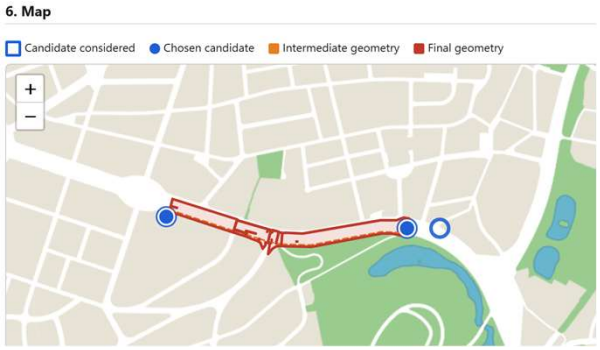
Candidates, chosen place and final geometry on a map

4 · Adjust

Tune the LLM prompt, scoring thresholds and the SQL builder

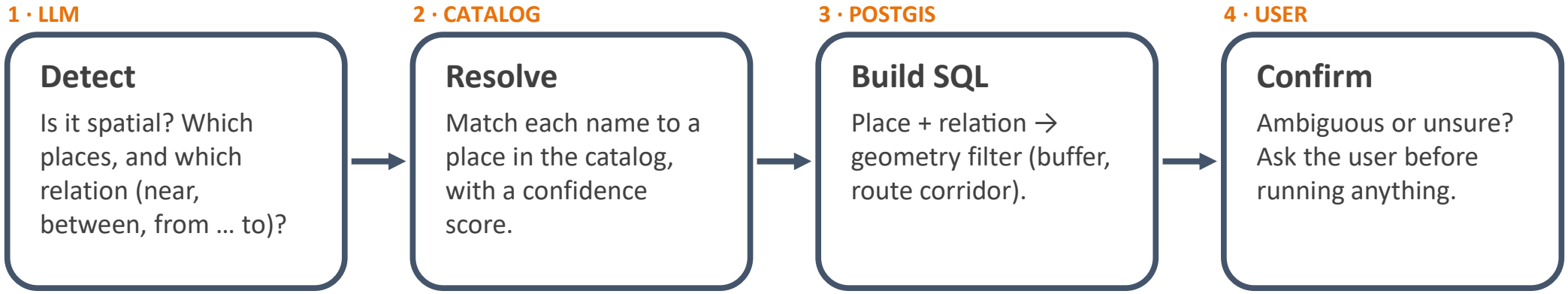
repeat until the prompts give the right place





How it works

One question travels through four modules



Exposed as MCP tools: geo_query · resolve_with_choice · resolve_place · resolve_address_tool · geometries_to_geojson

Fed by: UrbIS + STIB stops and lines + OSM + railway lines → PostGIS, synced regularly

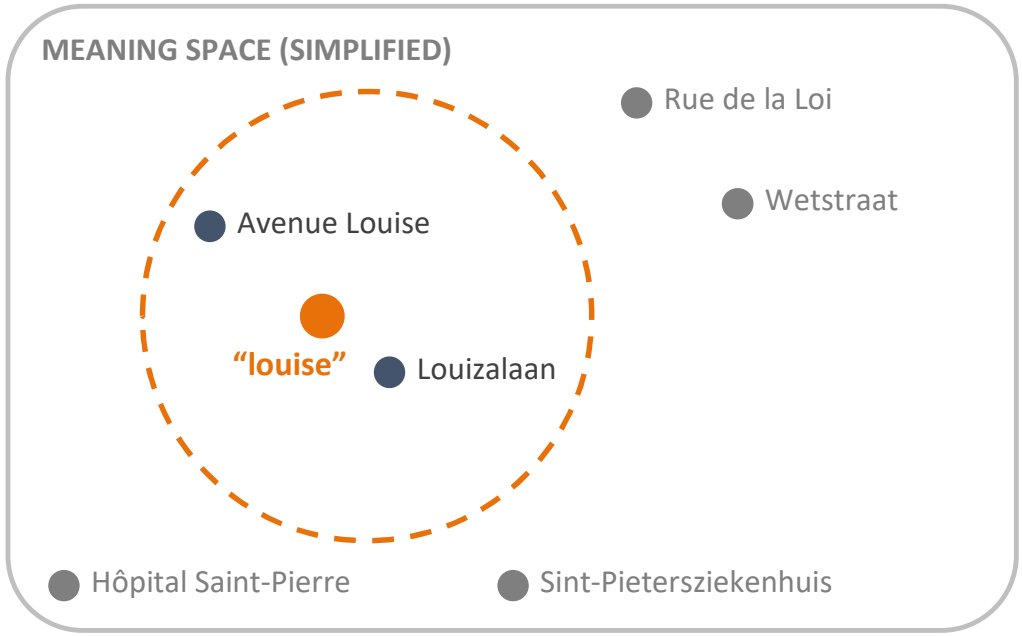
How it works

People type “Louise”, not “Avenue Louise” — names must match loosely

1. **Text match**
   pg_trgm: typos, abbreviations, shortened names

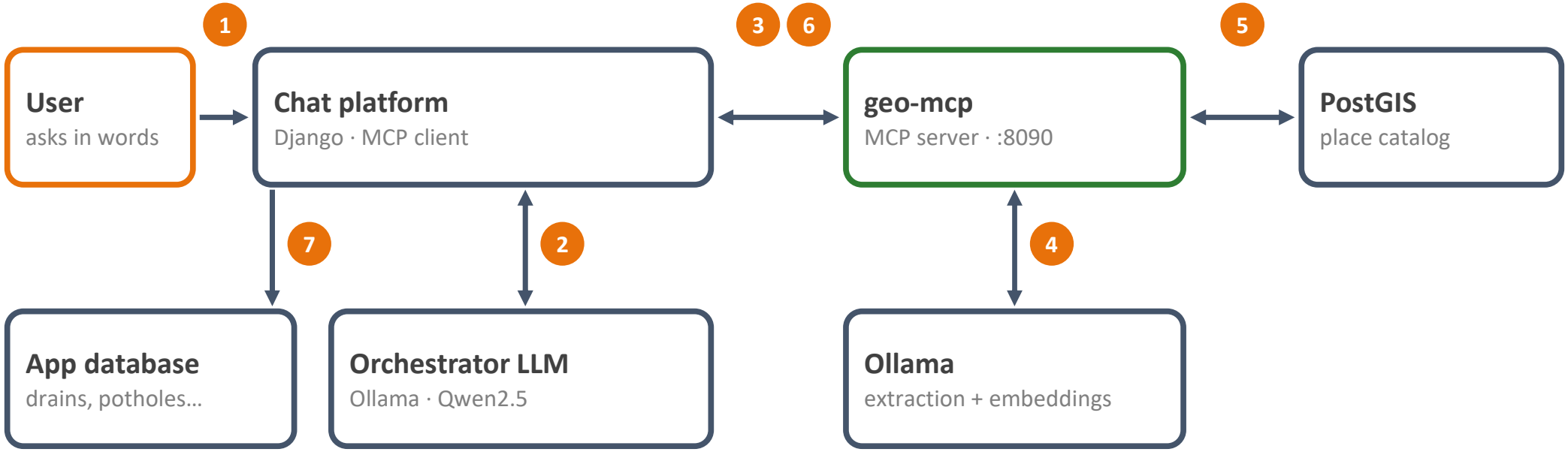
2. **Vector match**
   pgvector + bge-m3: if the text match is weak, compare meaning instead of letters

Each place name becomes 1024 numbers (FR / NL / EN). Similar names sit close together.



How it works

The chatbot calls geo-mcp, then runs the returned SQL itself



1 ask · 2 LLM picks the tool · 3 call geo_query · 4 extract + embed · 5 catalog lookup · 6 geometry comes back · 7 geometry is used by the app DB

Please help me

Your turn to try now:

Test it

Throw your own questions at it,
in French, Dutch or English.
Use a Brussels context.

Tell me where it fails

Wrong place, wrong road,
missing name? Every miss makes
the next version better.

THE CODE IS ON GITHUB

https://github.com/rcappellegob/geomcp